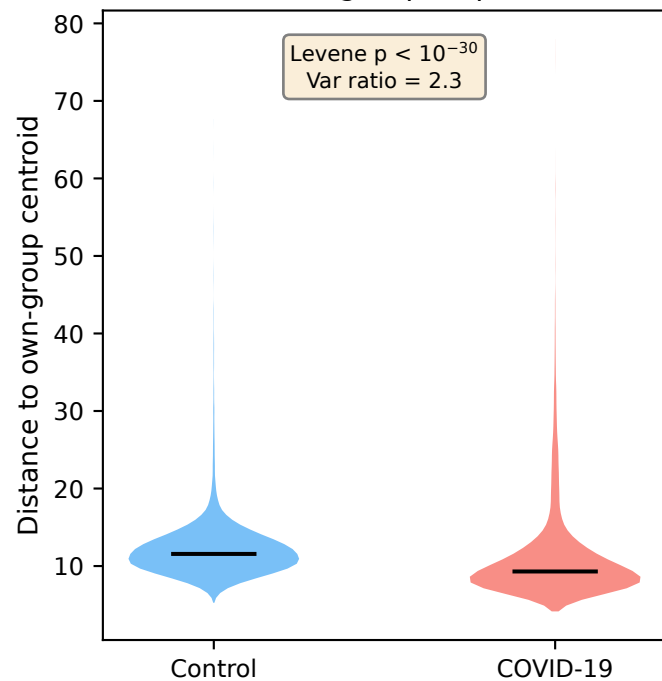
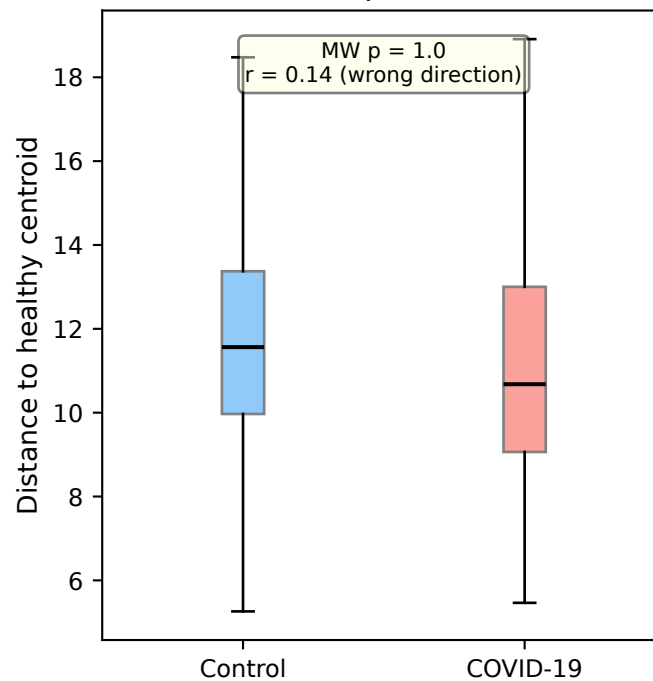


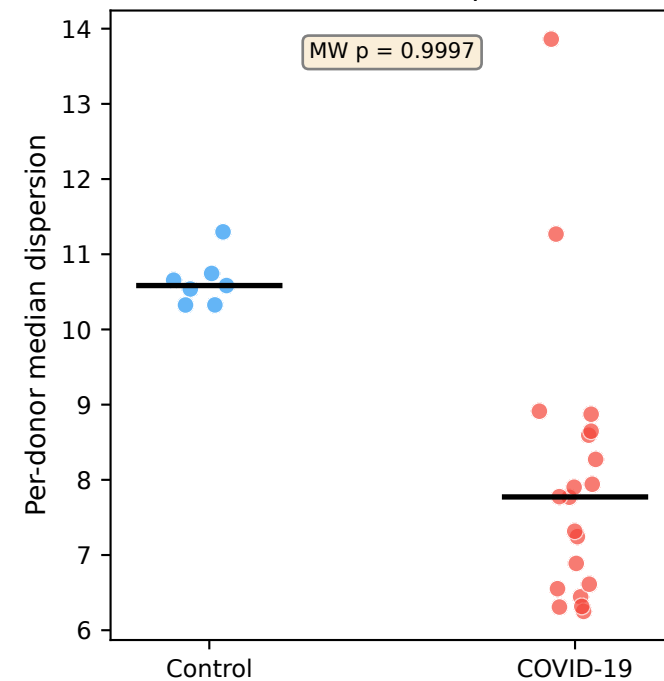
(a) Within-group dispersion



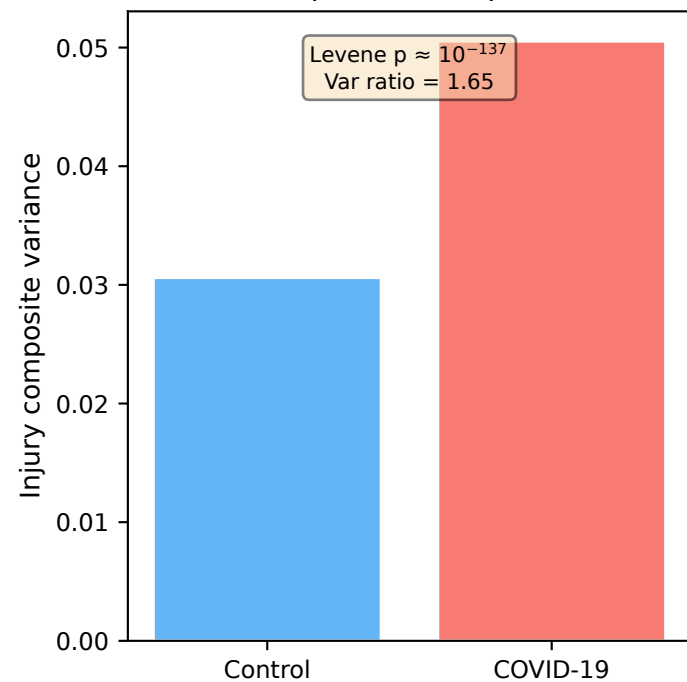
(b) Centroid displacement (FAILS)



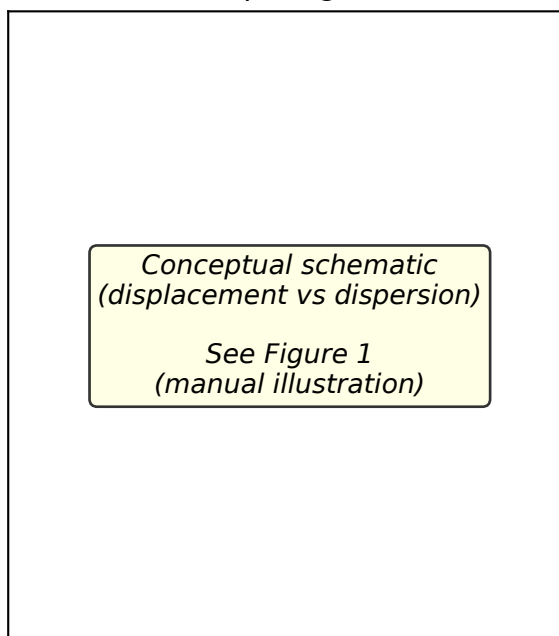
(c) Donor-level dispersion



(d) Replication dispersion



(e) Competing models



(f) Summary

Test	Primary	Replication
Displacement (centroid shift)	$p = 1.0$ (FAILS)	$p = 1.0$ (FAILS)
Dispersion (Levene's)	$p < 10^{-30}$ VR = 2.3	$p \approx 10^{-137}$ VR = 1.65
Pseudotime shift	$p = 0.001$ +0.048	$p = 0.002$ (donor-level)